

Qualification Pack



Refrigeration and Air Conditioning

QP Code: ELE/Q3127

Version: 2.0

NSQF Level: 4.5

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ELE/Q3127: Refrigeration and Air Conditioning

Brief Job Description

A Refrigeration and Air Conditioning Technician operates in diverse environments including commercial, residential, industrial, and transportation sectors ensuring the efficient functioning of systems that regulate temperature and humidity. Their role encompasses planning, designing, installing, testing, commissioning, and maintaining refrigeration and air conditioning systems. They handle a broad range of equipment and services, which are essential for preserving quality in areas such as food storage and ensuring comfort in buildings.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ELE/N3176: Work organization and management \(Refrigeration and Air Conditioning\)](#)
2. [ELE/N3177: Communication and interpersonal skills \(Refrigeration and Air Conditioning\)](#)
3. [ELE/N3178: Design refrigeration and air conditioning systems \(Refrigeration and Air Conditioning \)](#)
4. [ELE/N3179: Installation and maintenance of refrigeration and air conditioning \(Refrigeration and Air Conditioning\)](#)
5. [ELE/N3180: Commission refrigeration and air conditioning systems \(Refrigeration and Air Conditioning\)](#)
6. [ELE/N3181: Fault finding refrigeration and air conditioning systems \(Refrigeration and Air Conditioning\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	After Sales Service

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Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7421.0401
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	2.0
Reference code on NQR	QM-4.5-EH-05197-2026-V1-ESSCI
NQR Version	2

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ELE/N3176: Work organization and management (Refrigeration and Air Conditioning)

Description

This NOS unit is about proper training, the correct use of equipment, and adherence to safety protocols in manufacturing settings.

Scope

The scope covers the following :

- Principles and Applications of Safe Working
- Purposes, Uses, Care, and Maintenance of Equipment and Materials
- Environmental and Safety Principles in Good Housekeeping
- Principles of Team Working and Their Applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand and apply industry health and safety standards.
- PC2.** Identify hazards and implement control measures for workforce and public safety.
- PC3.** Perform risk assessments and prepare method statements.
- PC4.** Apply safe manual handling techniques, including mechanical lifting aids.
- PC5.** Select and correctly use personal protective equipment (PPE) for all activities.
- PC6.** Safely use access equipment, including ladders and mobile towers.
- PC7.** Apply electrical safety procedures while using tools and equipment.
- PC8.** Handle refrigerant and heat-producing equipment safely.
- PC9.** Maintain tidy and safe work areas, applying good housekeeping principles.
- PC10.** Respond appropriately to accidents and emergencies, including first aid for minor and major injuries.
- PC11.** Plan work efficiently within time and energy constraints, minimizing environmental impact.
- PC12.** Apply environmental protection, waste reduction, and safe disposal practices.
- PC13.** Incorporate energy conservation and smart technology practices in system design and maintenance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of preparing technical drawings and specifications as per engineering instructions.
- KU2.** Understanding of schematic capture and PCB layout design tools.
- KU3.** Understanding of schematic capture and PCB layout design tools.
- KU4.** Knowledge of CAD tools for electronic drawings across various systems.

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KU5. Understanding of professional reporting formats and documentation practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to prepare accurate technical drawings and specifications.
- GS2.** Ability to prepare accurate technical drawings and specifications.
- GS3.** Skill in creating schematic circuits using design software.
- GS4.** Ability to produce electronic drawings using CAD tools.
- GS5.** Ability to perform and record calculations with proper units.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Understand and apply industry health and safety standards.	-	-	-	-
PC2. Identify hazards and implement control measures for workforce and public safety.	-	-	-	-
PC3. Perform risk assessments and prepare method statements.	-	-	-	-
PC4. Apply safe manual handling techniques, including mechanical lifting aids.	-	1	-	-
PC5. Select and correctly use personal protective equipment (PPE) for all activities.	-	1	-	-
PC6. Safely use access equipment, including ladders and mobile towers.	-	1	-	-
PC7. Apply electrical safety procedures while using tools and equipment.	-	1	-	-
PC8. Handle refrigerant and heat-producing equipment safely.	-	1	-	-
PC9. Maintain tidy and safe work areas, applying good housekeeping principles.	-	1	-	-
PC10. Respond appropriately to accidents and emergencies, including first aid for minor and major injuries.	-	1	-	-
PC11. Plan work efficiently within time and energy constraints, minimizing environmental impact.	-	1	-	-
PC12. Apply environmental protection, waste reduction, and safe disposal practices.	-	1	-	-
PC13. Incorporate energy conservation and smart technology practices in system design and maintenance.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3176
NOS Name	Work organization and management (Refrigeration and Air Conditioning)
Sector	Electronics
Sub-Sector	
Occupation	After Sales Service
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N3177: Communication and interpersonal skills (Refrigeration and Air Conditioning)

Description

This NOS unit is about the communication and interpersonal skills.

Scope

The scope covers the following :

- Range and Purposes of Documentation and Publications in Electronic Forms
- Technical Language Associated with the Skill and Technology
- Standards for Routine and Exception Reporting in Oral and Electronic Form
- Required Standards for Communicating with Clients, Team Members, and Others

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the roles of members in construction teams and the building services industry, and collaborate effectively with related trades to complete tasks.
- PC2.** Use information sources and search methods to obtain specific and general technical data, specifications, and guidance.
- PC3.** Read, interpret, and extract technical data and instructions from manuals, documentation, and reports, including those generated by measuring equipment.
- PC4.** Communicate using the technical language of HVAC&R clearly and effectively through oral, written, and electronic means, adhering to standard reporting formats.
- PC5.** Apply required standards for health, safety, environment, and customer service, and respond appropriately to customer needs both directly and indirectly.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of roles and responsibilities within construction teams and the building services industry.
- KU2.** Knowledge of roles and responsibilities within construction teams and the building services industry.
- KU3.** Understanding of information sources and research methods for obtaining technical data and specifications.
- KU4.** Knowledge of interpreting technical manuals, documents, and measurement reports.
- KU5.** Understanding of HVAC&R technical language and standard communication formats.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Knowledge of health, safety, environmental, and customer service standards.
- GS2.** Ability to collaborate effectively with team members and related trades.
- GS3.** Skill in sourcing and utilizing technical information and guidance.
- GS4.** Ability to read, interpret, and extract relevant technical data accurately.
- GS5.** Skill in communicating technical information clearly in oral, written, and electronic forms.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Understand the roles of members in construction teams and the building services industry, and collaborate effectively with related trades to complete tasks.	-	1	-	-
PC2. Use information sources and search methods to obtain specific and general technical data, specifications, and guidance.	-	1	-	-
PC3. Read, interpret, and extract technical data and instructions from manuals, documentation, and reports, including those generated by measuring equipment.	-	1	-	-
PC4. Communicate using the technical language of HVAC&R clearly and effectively through oral, written, and electronic means, adhering to standard reporting formats.	-	1	-	-
PC5. Apply required standards for health, safety, environment, and customer service, and respond appropriately to customer needs both directly and indirectly.	-	1	-	-
NOS Total	-	5	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3177
NOS Name	Communication and interpersonal skills (Refrigeration and Air Conditioning)
Sector	Electronics
Sub-Sector	
Occupation	After Sales Service
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N3178: Design refrigeration and air conditioning systems (Refrigeration and Air Conditioning)

Description

This NOS unit is about evaluating system purposes, feasibility, and material properties while applying specialized knowledge in refrigeration, air conditioning, and related calculations.

Scope

The scope covers the following :

- Appraise the Purposes of Required Systems
- Appraise the Feasibility of Locating Systems Within Designated Areas
- Calculate Relative Densities of Refrigerants to Air and Water
- Apply Specialist Knowledge of the Principal Applications and Detailed Properties of Solid Materials

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand standard units of measurement, properties of materials and fluids, and relevant directives applicable to HVAC&R systems.
- PC2.** Understand the principles and interactions of energy, heat, power, force, and pressure as applied to HVAC&R systems.
- PC3.** Understand electricity, control circuits, and the types and uses of electrical cables and devices for HVAC&R applications.
- PC4.** Understand refrigeration and heat pump cycles, condensate drainage, and secondary refrigerant circuits.
- PC5.** Understand the principles, conventions, and range of specifications and drawings used in HVAC&R system design.
- PC6.** Appraise the purposes and feasibility of HVAC&R systems within designated spaces, considering environmental and operational constraints.
- PC7.** Apply specialist knowledge to calculate heat, energy, power, force, pressure, electrical parameters, and relative densities of refrigerants, fluids, and materials.
- PC8.** Design efficient refrigeration and air conditioning systems, selecting appropriate refrigerants, components, joining methods, and heat exchangers.
- PC9.** Produce technical drawings, specifications, and system layouts using standard conventions, symbols, and available drawing tools.
- PC10.** Estimate costs and budgets, select equipment and materials based on performance, environmental, and financial criteria, and adjust designs to maintain budget compliance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of standard measurement units, material properties, and HVAC&R directives.

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- KU2.** Knowledge of standard measurement units, material properties, and HVAC&R directives.
- KU3.** Understanding of energy, heat, power, force, and pressure principles in HVAC&R systems.
- KU4.** Knowledge of electrical systems, control circuits, and related components in HVAC&R applications.
- KU5.** Knowledge of HVAC&R design drawings, specifications, and conventions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to apply measurement standards and material knowledge in HVAC&R work.
- GS2.** Ability to apply measurement standards and material knowledge in HVAC&R work.
- GS3.** Skill in analysing thermal and mechanical system parameters.
- GS4.** Ability to understand and work with electrical systems and control circuits.
- GS5.** Skill in applying refrigeration and heat pump principles in system operations.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Understand standard units of measurement, properties of materials and fluids, and relevant directives applicable to HVAC&R systems.	-	1	-	-
PC2. Understand the principles and interactions of energy, heat, power, force, and pressure as applied to HVAC&R systems.	-	1	-	-
PC3. Understand electricity, control circuits, and the types and uses of electrical cables and devices for HVAC&R applications.	-	1	-	-
PC4. Understand refrigeration and heat pump cycles, condensate drainage, and secondary refrigerant circuits.	-	1	-	-
PC5. Understand the principles, conventions, and range of specifications and drawings used in HVAC&R system design.	-	1	-	-
PC6. Appraise the purposes and feasibility of HVAC&R systems within designated spaces, considering environmental and operational constraints.	-	1	-	-
PC7. Apply specialist knowledge to calculate heat, energy, power, force, pressure, electrical parameters, and relative densities of refrigerants, fluids, and materials.	-	1	-	-
PC8. Design efficient refrigeration and air conditioning systems, selecting appropriate refrigerants, components, joining methods, and heat exchangers.	-	1	-	-
PC9. Produce technical drawings, specifications, and system layouts using standard conventions, symbols, and available drawing tools.	-	1	-	-
PC10. Estimate costs and budgets, select equipment and materials based on performance, environmental, and financial criteria, and adjust designs to maintain budget compliance.	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3178
NOS Name	Design refrigeration and air conditioning systems (Refrigeration and Air Conditioning)
Sector	Electronics
Sub-Sector	
Occupation	After Sales Service
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N3179: Installation and maintenance of refrigeration and air conditioning (Refrigeration and Air Conditioning)

Description

This NOS unit is about interpreting plans for piping and electrical routes, safely handling materials and equipment in the RAC industry, and performing installations, jointing, and testing of refrigeration and air conditioning systems.

Scope

The scope covers the following :

- Interpret Diagrams, Plans, and Specifications to Determine Appropriate Piping and Electricity Routes
- Work Safely with Hot Gas Heating Equipment in the RAC Industry
- Prepare and, If Needed, Improve the Area and Surfaces Which the Installation Depends Upon
- Itemize the Required Tools, Components, and Materials for Installations

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand and apply specific health and safety requirements for fitting, fixing, servicing, maintaining, and decommissioning HVAC&R systems.
- PC2.** Understand the working principles, layouts, and operational characteristics of HVAC&R and heat pump systems.
- PC3.** Interpret diagrams, plans, and specifications to determine appropriate piping and electrical routes.
- PC4.** Apply procedures for fitting, fixing, and testing materials, equipment, and components safely and accurately.
- PC5.** Prepare and, if necessary, improve installation areas and surfaces to meet operational requirements.
- PC6.** Itemize and organize required tools, components, and materials for installation tasks.
- PC7.** Take and transfer measurements and angles from drawings to surfaces and piping materials accurately.
- PC8.** Identify, check, and safely use various types of gases and equipment used for joining materials in the HVAC&R industry.
- PC9.** Safely drain oil and recover refrigerants, including HC, HFC, HFO, and natural refrigerants.
- PC10.** Join similar and dissimilar materials using permanent (brazed and compression/crimped) and accessible (flared) jointing methods according to industry standards.
- PC11.** Fabricate and install mechanical materials and components according to drawings and specifications.
- PC12.** Install electrical materials, components, and control devices as per drawings and specifications.
- PC13.** Install ancillary components and systems such as condensate drainage and leak detection systems.

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- PC14.** Use tools and equipment to test the strength, tightness, and leak-free condition of refrigeration systems or components (e.g., per ISO 5149.2:2014).
- PC15.** Use tools and equipment to evacuate moisture and non-condensable gases, charge and recover refrigerants (HC, HFC, HFO, and natural refrigerants), ensuring systems are dry and leak-free.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of health and safety requirements for HVAC&R installation, servicing, and decommissioning.
- KU2.** Understanding of working principles and operational characteristics of HVAC&R and heat pump systems.
- KU3.** Knowledge of interpreting diagrams, plans, and specifications for piping and electrical routing.
- KU4.** Understanding of procedures for fitting, fixing, and testing materials and components
- KU5.** Understanding of procedures for fitting, fixing, and testing materials and components

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Knowledge of site preparation and surface readiness for installation.
- GS2.** Ability to apply safety requirements during installation and servicing activities.
- GS3.** Skill in understanding and working with HVAC&R system operations.
- GS4.** Ability to interpret drawings and determine installation routes accurately.
- GS5.** Ability to prepare and improve installation areas effectively.
- GS6.** Ability to prepare and improve installation areas effectively.
- GS7.** Skill in organizing tools, materials, and components for tasks.
- GS8.** Ability to take and transfer measurements accurately.
- GS9.** Skill in handling gases and using joining equipment safely.
- GS10.** Ability to drain oil and recover refrigerants safely.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	30	-	-
PC1. Understand and apply specific health and safety requirements for fitting, fixing, servicing, maintaining, and decommissioning HVAC&R systems.	-	2	-	-
PC2. Understand the working principles, layouts, and operational characteristics of HVAC&R and heat pump systems.	-	2	-	-
PC3. Interpret diagrams, plans, and specifications to determine appropriate piping and electrical routes.	-	2	-	-
PC4. Apply procedures for fitting, fixing, and testing materials, equipment, and components safely and accurately.	-	2	-	-
PC5. Prepare and, if necessary, improve installation areas and surfaces to meet operational requirements.	-	2	-	-
PC6. Itemize and organize required tools, components, and materials for installation tasks.	-	2	-	-
PC7. Take and transfer measurements and angles from drawings to surfaces and piping materials accurately.	-	2	-	-
PC8. Identify, check, and safely use various types of gases and equipment used for joining materials in the HVAC&R industry.	-	2	-	-
PC9. Safely drain oil and recover refrigerants, including HC, HFC, HFO, and natural refrigerants.	-	2	-	-
PC10. Join similar and dissimilar materials using permanent (brazed and compression/crimped) and accessible (flared) joining methods according to industry standards.	-	2	-	-
PC11. Fabricate and install mechanical materials and components according to drawings and specifications.	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Install electrical materials, components, and control devices as per drawings and specifications.	-	2	-	-
PC13. Install ancillary components and systems such as condensate drainage and leak detection systems.	-	2	-	-
PC14. Use tools and equipment to test the strength, tightness, and leak-free condition of refrigeration systems or components (e.g., per ISO 5149.2:2014).	-	2	-	-
PC15. Use tools and equipment to evacuate moisture and non-condensable gases, charge and recover refrigerants (HC, HFC, HFO, and natural refrigerants), ensuring systems are dry and leak-free.	-	2	-	-
NOS Total	-	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3179
NOS Name	Installation and maintenance of refrigeration and air conditioning (Refrigeration and Air Conditioning)
Sector	Electronics
Sub-Sector	
Occupation	After Sales Service
NSQF Level	4.5
Credits	5
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N3180: Commission refrigeration and air conditioning systems (Refrigeration and Air Conditioning)

Description

This NOS unit is about charging RAC systems with the correct refrigerant, assessing their performance, identifying issues, and adjusting controls for optimal operation.

Scope

The scope covers the following :

- Charge Systems with the Correct Type and Quantity of Refrigerant for Efficient Operation Without Leaking Refrigerant to the Environment
- Examine Systems for Leakage After Commissioning, Using Direct or Indirect Measuring Methods
- Assess Refrigeration Systems for Correct Operation
- Assess Air Conditioning Systems for Correct Operation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand how to interpret the design parameters of HVAC&R systems for commissioning.
- PC2.** Apply safety standards for handling and applying refrigerants.
- PC3.** Follow safe procedures for connecting and applying electrical supply to HVAC&R systems.
- PC4.** Set up safety controls and devices to meet design requirements.
- PC5.** Ensure the integrity and correct functioning of HVAC&R systems.
- PC6.** Charge systems with the correct type and quantity of refrigerant for efficient and leak-free operation.
- PC7.** Examine systems for leakage after commissioning using direct or indirect measuring methods (e.g., ISO 5149.2:2014).
- PC8.** Assess refrigeration, air conditioning, secondary refrigerant, and cooling systems for correct operation.
- PC9.** Assess the electrical installation of HVAC&R systems for correct operation and compliance.
- PC10.** Solve operational problems affecting the safety and efficiency of RAC systems.
- PC11.** Adjust refrigerant controls, flow devices, and electrical/electronic controls for optimal system performance.
- PC12.** Balance air distribution systems, measure and record operating parameters, and ensure availability of essential client documentation.
- PC13.** Provide client training, demonstrate safe functioning, and hand over systems while responding to all relevant queries.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Knowledge of HVAC&R design parameters and commissioning requirements.
- KU2.** Understanding of safety standards for handling and applying refrigerants.
- KU3.** Understanding of safety standards for handling and applying refrigerants.
- KU4.** Knowledge of safe electrical connection procedures for HVAC&R systems.
- KU5.** Knowledge of system integrity and performance verification methods.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to interpret design parameters for effective commissioning.
- GS2.** Ability to interpret design parameters for effective commissioning.
- GS3.** Skill in applying safety standards while handling refrigerants.
- GS4.** Ability to connect and manage electrical supply safely.
- GS5.** Skill in setting up and verifying safety controls and devices.
- GS6.** Ability to ensure system integrity and proper functioning.
- GS7.** Ability to detect and assess system leaks accurately.
- GS8.** Ability to detect and assess system leaks accurately.
- GS9.** Skill in evaluating system performance and operational efficiency.
- GS10.** Skill in troubleshooting and resolving operational issues.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	25	-	-
PC1. Understand how to interpret the design parameters of HVAC&R systems for commissioning.	-	2	-	-
PC2. Apply safety standards for handling and applying refrigerants.	-	2	-	-
PC3. Follow safe procedures for connecting and applying electrical supply to HVAC&R systems.	-	2	-	-
PC4. Set up safety controls and devices to meet design requirements.	-	2	-	-
PC5. Ensure the integrity and correct functioning of HVAC&R systems.	-	2	-	-
PC6. Charge systems with the correct type and quantity of refrigerant for efficient and leak-free operation.	-	2	-	-
PC7. Examine systems for leakage after commissioning using direct or indirect measuring methods (e.g., ISO 5149.2:2014).	-	2	-	-
PC8. Assess refrigeration, air conditioning, secondary refrigerant, and cooling systems for correct operation.	-	2	-	-
PC9. Assess the electrical installation of HVAC&R systems for correct operation and compliance.	-	2	-	-
PC10. Solve operational problems affecting the safety and efficiency of RAC systems.	-	2	-	-
PC11. Adjust refrigerant controls, flow devices, and electrical/electronic controls for optimal system performance.	-	2	-	-
PC12. Balance air distribution systems, measure and record operating parameters, and ensure availability of essential client documentation.	-	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Provide client training, demonstrate safe functioning, and hand over systems while responding to all relevant queries.	-	1	-	-
NOS Total	-	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3180
NOS Name	Commission refrigeration and air conditioning systems (Refrigeration and Air Conditioning)
Sector	Electronics
Sub-Sector	
Occupation	After Sales Service
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N3181: Fault finding refrigeration and air conditioning systems (Refrigeration and Air Conditioning)

Description

This NOS unit is about inspecting, testing, and diagnosing issues in electrically operated RAC systems, ensuring safety during fault rectification and refrigerant management.

Scope

The scope covers the following :

- Inspect and Test Electrically Operated RAC System Components
- Solve Problems Affecting the Safe and Effective Operation of RAC Systems
- Safely Diagnose and Rectify Faults in Electrically Operated RAC Services and Components
- Prepare and Safeguard Areas of Work and Their Surroundings

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand electrical standards applicable to the RAC industry and inspection/testing requirements for electrically operated RAC services and components.
- PC2.** Apply safe procedures for diagnosing and rectifying faults in electrically operated RAC services and components
- PC3.** Understand the layout and features of electrical circuits in RAC systems and the importance of original design and operational parameters.
- PC4.** Carry out risk assessments and safely isolate electrical systems prior to troubleshooting or maintenance.
- PC5.** Inspect, test, and assess electrically operated HVAC&R components and systems for correct operation and integrity.
- PC6.** Solve problems affecting the safe and effective operation of RAC systems.
- PC7.** Reclaim and handle refrigerants (HC, HFC, HFO, and natural refrigerants like Ammonia/CO₂) safely without environmental leakage.
- PC8.** Replace faulty components, drain/refill compressor lubricants, and restore refrigerant system integrity.
- PC9.** Restore work areas and surroundings to their prior state and ensure safety compliance.
- PC10.** Explain, advise, and report on findings, actions, and matters requiring further attention to stakeholders or clients.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of electrical standards and inspection/testing requirements in the RAC industry.
- KU2.** Knowledge of electrical standards and inspection/testing requirements in the RAC industry.

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- KU3.** Understanding of safe fault diagnosis and rectification procedures for electrical systems.
- KU4.** Knowledge of electrical circuit layouts and operational parameters in RAC systems.
- KU5.** Understanding of risk assessment and electrical isolation procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to interpret and apply electrical standards in RAC systems.
- GS2.** Ability to interpret and apply electrical standards in RAC systems.
- GS3.** Skill in diagnosing and rectifying electrical faults safely.
- GS4.** Ability to understand and work with RAC electrical circuit layouts.
- GS5.** Skill in conducting risk assessments and isolating systems safely.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	20	-	-
PC1. Understand electrical standards applicable to the RAC industry and inspection/testing requirements for electrically operated RAC services and components.	-	2	-	-
PC2. Apply safe procedures for diagnosing and rectifying faults in electrically operated RAC services and components	-	2	-	-
PC3. Understand the layout and features of electrical circuits in RAC systems and the importance of original design and operational parameters.	-	2	-	-
PC4. Carry out risk assessments and safely isolate electrical systems prior to troubleshooting or maintenance.	-	2	-	-
PC5. Inspect, test, and assess electrically operated HVAC&R components and systems for correct operation and integrity.	-	2	-	-
PC6. Solve problems affecting the safe and effective operation of RAC systems.	-	2	-	-
PC7. Reclaim and handle refrigerants (HC, HFC, HFO, and natural refrigerants like Ammonia/CO ₂) safely without environmental leakage.	-	2	-	-
PC8. Replace faulty components, drain/refill compressor lubricants, and restore refrigerant system integrity.	-	2	-	-
PC9. Restore work areas and surroundings to their prior state and ensure safety compliance.	-	2	-	-
PC10. Explain, advise, and report on findings, actions, and matters requiring further attention to stakeholders or clients.	-	2	-	-
NOS Total	-	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N3181
NOS Name	Fault finding refrigeration and air conditioning systems (Refrigeration and Air Conditioning)
Sector	Electronics
Sub-Sector	
Occupation	After Sales Service
NSQF Level	4.5
Credits	3
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take

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subsequent assessment on the balance NOS's to pass the Qualification Pack.

6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N3176.Work organization and management (Refrigeration and Air Conditioning)	-	10	-	-	10	10
ELE/N3177.Communication and interpersonal skills (Refrigeration and Air Conditioning)	-	5	-	-	5	5
ELE/N3178.Design refrigeration and air conditioning systems (Refrigeration and Air Conditioning)	-	10	-	-	10	10
ELE/N3179.Installation and maintenance of refrigeration and air conditioning (Refrigeration and Air Conditioning)	-	30	-	-	30	30
ELE/N3180.Commission refrigeration and air conditioning systems (Refrigeration and Air Conditioning)	-	25	-	-	25	25
ELE/N3181.Fault finding refrigeration and air conditioning systems (Refrigeration and Air Conditioning)	-	20	-	-	20	20

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	0	100	-	-	100	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.